Per-Class Threshold Calibration for Training-Free Open-Vocabulary Semantic Segmentation of Remote Sensing Imagery

Mid-Project Review

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డిజైన్ అండ్ మాన్యుఫ్యాక్చరింగ్, కర్నూలు

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Presentation Outline

1. Introduction

Open-vocabulary semantic segmentation of aerial imagery. The input is an image together with a list of class names given as plain words; the output is a class label for every pixel. The setting is **training-free**: no network weights are learned, and the class list may change at inference without retraining.

Baseline. SegEarth-OV3 [1] applies SAM 3 [2] to remote sensing. For each class prompt, SAM 3 emits a presence score, a dense semantic map and a set of instance masks; SegEarth-OV3 fuses these and applies a single confidence threshold τ, tuned per dataset. Pixels below τ are assigned to a catch-all class. Reproduced on our hardware at 47.38 mIoU against a published 47.4.

Contribution

A single confidence threshold is the wrong *shape* for a multi-class pipeline. We fit **one threshold per class**, and **one scale per class applied before the argmax**, using the same supervision budget the baseline already spends tuning its single τ. **No network weights are trained.**

2. Motivation — the baseline discards land cover

Measured on LoveDA validation at SegEarth-OV3's own published $\tau = 0.5$:

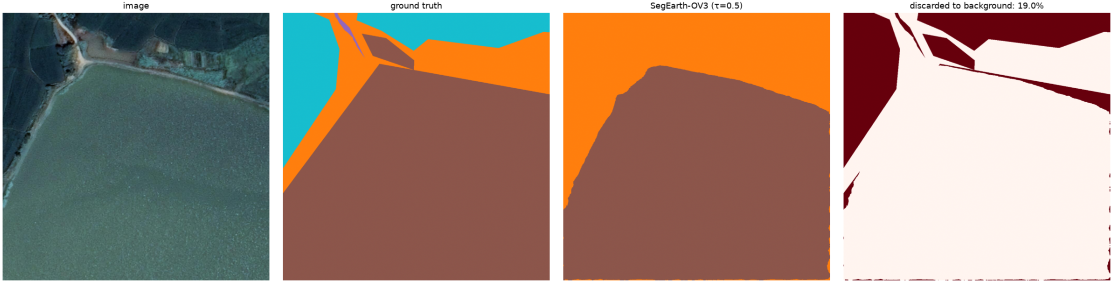
- **29.68%** of all real-class pixels are assigned to the catch-all class `background` — **323 million pixels**.
- Discard exceeds real-class confusion by **3:1**. The dominant error mode is abstention rather than misclassification.
- `water`: precision **89.5%**, recall **54.7%**. The class is correct nine times in ten when predicted, yet recovers barely half of the water present.

Cost of global remedies

Lower τ to 0.1	−5.54 mIoU; 1.73 incorrect pixels per correct one
Remove presence gating	−11.97 mIoU
No intervention	29.68% of land cover discarded

A correction that operates uniformly across classes therefore cannot recover the residual. **The correction must be per-class.**

2. Motivation — the discard, visualised



A LoveDA validation tile. Left to right: **input image**, **ground truth**, **SegEarth-OV3 prediction at** $\tau = 0.5$, and the **pixels assigned to `background`**. Entire water regions are discarded — not misclassified, but declared unknown — because their confidence falls below a threshold tuned for the dataset as a whole rather than for `water`.

2. Application domains

The setting is intended for deployments in which the class list is not fixed in advance and labelled data for the target region does not exist.

- **Disaster response.** Flood or damage extent from UAV imagery acquired within hours of an event, where no annotated data exists for the affected district and there is no time to train a model.
- **Infrastructure monitoring.** Classes such as `solar panel` or `transmission line` appear in no land-cover benchmark, yet can be supplied to the pipeline as words.
- **Settlement mapping.** Vocabularies such as `tin roof`, `unpaved road` and `water tank` change between surveys and between regions.

Deployment cost

The proposed correction requires **∼200 labelled tiles** and **no gradient steps**. Adapting the same pipeline by fine-tuning would require labelled data in the thousands of tiles, a training run, and would forfeit the training-free property.

3. Problem statement — formulation

Let $\mathcal{C} = \{1, \ldots, N\}$ denote the vocabulary and $s_c(p) \in [0, 1]$ the score the pipeline assigns to class c at pixel p. Let b denote the catch-all class.

Baseline decision rule, with a single global threshold τ:

$$\hat{y}(p) = \begin{cases} \arg\max_c s_c(p), & \text{if } \max_c s_c(p) \geq \tau \\ b, & \text{otherwise} \end{cases}$$

Proposed rule, with parameters fitted on a calibration set of ∼200 labelled tiles:

$$\hat{y}(p) = \begin{cases} c^\star, & \text{if } s_{c^\star}(p) \geq \boldsymbol{\tau}_{\boldsymbol{c}^\star} \\ b, & \text{otherwise} \end{cases} \qquad c^\star = \arg\max_c \ \boldsymbol{w}_{\boldsymbol{c}}\, s_c(p)$$

Objective

$(\boldsymbol{w}, \boldsymbol{\tau}) = \arg\max \ \mathrm{mIoU}_{\mathrm{real}}$, optimised by coordinate ascent. **No network weights are modified.** The threshold is applied to the *raw* score, so $\boldsymbol{w}$ influences only the `argmax`.

3. Completeness of the per-class threshold family

Proposition. For a fixed `argmax`, any strictly increasing per-class recalibration g_c composed with a global threshold τ is equivalent to a per-class threshold vector:

$$g_c(s) \;\geq\; \tau \quad\iff\quad s \;\geq\; g_c^{-1}(\tau) \;=\; \tau_c.$$

- **Per-class thresholds therefore exhaust every correction reachable by rescaling scores after the `argmax`**; no member of that family can outperform them.
- Scaling applied *before* the `argmax` alters which class wins, and so constitutes a **strictly larger family**.

Empirical consequence

On LoveDA, **6.0%** of catch-all assignments are pixels for which the catch-all class won the `argmax` at $s \geq \tau$. A threshold cannot alter an `argmax`, so this residual is unreachable by the first parameter set and motivates the second.

4. Literature survey

No.	Work	Year	Contribution and relevance
1	**CLIP** [3]	2021	Image–text contrastive pretraining; origin of the open-vocabulary paradigm.
2	**MaskCLIP** [4]	2022	First extraction of dense predictions from CLIP by removing pooling.
3	**GroupViT** [5]	2022	Learns segment grouping from text supervision alone.
4	**SCLIP** [6]	2023	Correlative self-attention; the codebase from which SegEarth-OV3 derives.
5	**SAM** [7]	2023	Promptable class-agnostic segmentation without a fixed label set.
6	**ClearCLIP** [8]	2024	Identifies which CLIP components degrade dense prediction.
7	**ProxyCLIP** [9]	2024	Self-supervised features as structural guidance for segmentation.
8	**SegEarth-OV** [10]	2025	Feature upsampling and global bias subtraction for remote sensing; establishes the evaluation protocol.
9	**SAM 3** [2]	2025	Presence, semantic and instance heads unified in one promptable model.
10	**SegEarth-OV3** [1]	2025	SAM 3 for remote sensing: dual-head fusion and presence gating. **Baseline of this work.**
11	**ConInfer** [11]	2026	Gaussian mixture over self-supervised patch features fused

5. Research gaps

Gap 1 — the decision rule is unexamined

Every method in this family terminates in an `argmax` followed by a single global threshold, tuned per dataset. Research effort is directed at improving the *scores*; whether the *decision rule* applied to them has the appropriate form has not been studied.

Gap 2 — the discarded residual is not quantified

Published work reports mIoU. It does not report what fraction of real land cover is assigned to the catch-all class, nor whether that fraction is recoverable.

Gap 3 — catch-all classes distort the reported metric

mIoU is an *unweighted* mean over classes, so a catch-all class contributes $1/N$ of the score irrespective of its semantic content. A method may therefore appear to improve or degrade segmentation while land-cover quality is unchanged.

6. Objectives

Completed

1. **Reproduce** SegEarth-OV3 on local hardware, so that every subsequent measurement is made against a baseline we control. 47.38 vs published 47.4
2. **Quantify** the residual the baseline discards, and determine whether any global adjustment recovers it. 29.68%; no global adjustment does
3. **Design** a correction that trains no weights and consumes only the supervision budget the baseline already spends. two parameter sets

In progress

4. **Extend** the correction beyond a single vector per dataset: the fitted parameters are currently constant across tiles, and a scene-adaptive formulation remains to be investigated.
5. **Investigate** a per-class fusion of SAM 3's semantic and instance heads, which are presently combined by a fixed maximum.
6. **Broaden** the evaluation to further high-resolution benchmarks and establish the conditions under which the correction transfers.

7. Experimental setup

Datasets

	Tiles	GSD	Classes
LoveDA (val)	1669	0.3 m	7
ISPRS Potsdam	2016	5 cm	6
OpenEarthMap	384	0.25–0.5 m	9

Baselines and controls

- SegEarth-OV3, reproduced — the primary baseline
- Global τ sweep; presence-gating removal
- Per-class Otsu and eleven further label-free rules

Protocol

- Five-fold cross-validation. Calibration and evaluation tiles are **always disjoint**.
- The baseline is recomputed on the *same* held-out tiles, so both rules are scored on identical pixels.
- Both metrics are reported: full mIoU and mIoU excluding the catch-all class.
- Results are confirmed by the official evaluator, not by arithmetic on a cached statistic.

Validation gate

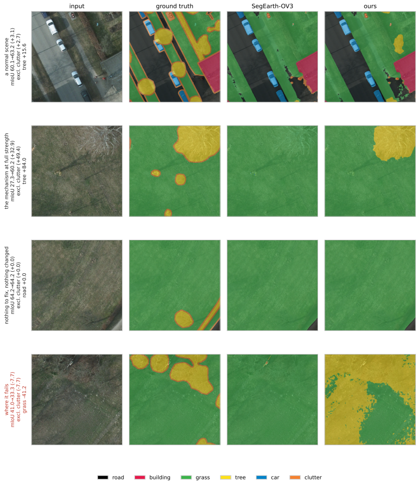
Every instrumented run must still reproduce **47.37 mIoU** and **29.68%** discard. A deviation indicates that the instrumentation altered model behaviour.

7. Results

	LoveDA 1669 tiles	Potsdam 2016 tiles	OpenEarthMap 384 tiles
SegEarth-OV3, reproduced	47.38	57.60	44.16
+ per-class thresholds	$+1.18 \pm 0.45$	$+0.59$	$+0.30$
+ per-class scaling	$+1.16 \pm 0.19$	$+4.86 \pm 0.35$	—
Total Δ mIoU	**+2.32**	**+5.67**	+0.30
Δ excluding catch-all	+1.36	+6.00	**+1.75**
Folds positive	5 / 5	5 / 5	—

- **Confirmed end to end.** On Potsdam the official evaluator gives 57.60 → 58.35 → 63.27 over 1816 held-out tiles, each stage within **0.05** mIoU of its predicted value.
- **Largest per-class change:** Potsdam `tree`, **37.92 → 59.09** IoU — a class with precision 93.34% and recall 38.63%, the asymmetry the method is designed to correct.
- **OpenEarthMap** is positive under the catch-all-excluded metric (+1.75) although full mIoU appears flat: its catch-all class contributes one ninth of the mean and offsets the gain.

7. Results — qualitative



Four ISPRS Potsdam tiles. Columns: input, ground truth, SegEarth-OV3, and the calibrated rule. **The network, its weights and the forward pass are identical across the two right-hand columns; only the decision rule differs.** The fourth row is a tile on which the calibrated rule loses 7.72 mIoU: the

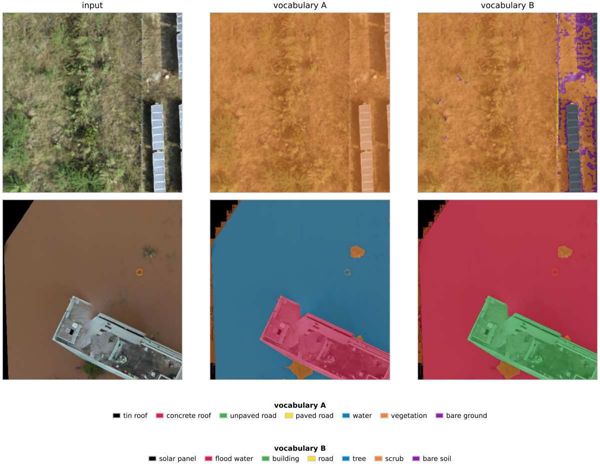
8. Application study — an unseen region and vocabulary

The correction was applied to 25 tiles cut from 5 openly licensed UAV scenes over India at 4–5 cm ground sample distance — a region present in none of the three benchmarks — under two class lists supplied as words at inference.

- The segmentation is stable under renaming. A flooded building is partitioned almost identically by both vocabularies: 78.6% `water` against 79.1% `flood water`, and 18.2% `concrete roof` against 18.3% `building`. The regions are recovered consistently; only the labels follow the supplied words.
- A class absent from the vocabulary is not recovered. 100.0% of a photovoltaic-array tile is assigned to a single class until `solar panel` is supplied, which then accounts for 4.6% — below the visible extent of the panels.

No ground truth exists for this imagery. The quantities above are per-class pixel shares and are not accuracy measurements.

8. Application study — one scene, two vocabularies



Indian UAV imagery at 4–5 cm. Identical weights and forward pass; only the supplied class list differs between the second and third columns. **Lower row:** a building in flood water, recovered consistently under both vocabularies. **Upper row:** a photovoltaic array, for which the first vocabulary provides no

Project timeline

Period	Activity	Status
Aug 2026	Environment, baseline reproduction, discard measurement	Completed
Aug 2026	Co-occurrence prior: constructed, evaluated, rejected	Completed
Aug–Sep 2026	Per-class thresholds; cross-validation and deployment	Completed
Sep 2026	Per-class scaling; end-to-end verification on two datasets	Completed
Sep 2026	Application study on unseen imagery	Completed
Oct–Dec 2026	Scene-adaptive formulation; head-fusion investigation	Ongoing
Oct–Dec 2026	Additional high-resolution benchmarks	Ongoing
Jan–Feb 2027	Manuscript preparation and internal review	Planned
Q1 2027	Journal submission	Planned

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Thank You